Markus Buchholz

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Research Interests

Autonomous systems, robotics and control theory; neuro-symbolic AI and world-model-grounded planning; real-time trajectory optimisation and model predictive control; multi-robot coordination and swarm intelligence; spatial AI and SLAM; physics-based simulation and digital twins for maritime, aerial, and defence platforms.

Education

Gdansk University of Technology **Jan 2010 – Dec 2012**
Ph.D. in Mechatronics
Research: Real-time algorithms for nonlinear robot control systems.

Gdansk University of Technology **Sep 1998 – Jun 2003**
M.Sc. in Electronics, Telecommunication, and Computer Science
Grade: A

Gdansk University of Technology **Mar 2002 – Mar 2004**
M.Sc. in Economy and Management
Grade: A (with distinction)

Professional Experience

Research Scientist in AI and Simulation Technology **Jan 2026 – present**
Norwegian Defence Research Establishment (FFI), Kjeller, Norway
Developing world-model-grounded AI architectures for autonomous tactical decision-making, integrated with high-fidelity defence-grade simulators. Research spans neuro-symbolic planning, LLM-based mission reasoning, and multi-agent coordination for military systems.

Postdoctoral Researcher in Underwater Robotics **Sep 2023 – Dec 2025**
Ocean Systems Lab, Heriot-Watt University, Edinburgh, Scotland
Research on cognitive autonomy for underwater robots: How can multi-agent autonomous systems perform diagnostics and maintenance of offshore wind farms? This includes investigating how autonomous underwater vehicles can detect and diagnose anomalies, adapt mission plans in response to changing conditions, and coordinate as teams rather than simply executing pre-programmed or scripted behaviours. Conceived and built UROSA and AURA, open-source decentralised multi-agent autonomy frameworks for embedded AUV deployment; first-author publications at IROS, ICRA and OCEANS, two journal articles in press (*Journal of Ocean Engineering*), and three manuscripts under review, two of them on world-model-grounded LLM planning for AUVs and ASVs. The research was validated on real systems: lead software architect for UNITE (ASV, tethered AUV, 7-DOF manipulator arm), designing the complete autonomous ROS 2 stack and delivering it to Fugro with a successful field demonstration. Co-developer on HUME multi-agent tracking (SeeByte); whole-body MPC with Crocoddyl and Pinocchio (RoMI Lab, Dr. Carlos Mastalli). Open-source contributor to the GazeboSim marine simulation community. Teaching Assistant, Robotics Systems Science ($\approx$30 sessions); supervised an MSc project (top grade), a BSc project (student admitted to a doctoral programme in Beijing), and a physical AUV build; managed laboratory operations and hardware demonstrations.

Senior Software Engineer – Robotics **Mar 2023 – Sep 2023**
Yaskawa Robotics, Germany

Developed and integrated ROS 2 software interfaces for Yaskawa industrial manipulators, contributing to the MotoROS2 open-source project - the official ROS 2 driver for Yaskawa robots. Designed motion planning and real-time trajectory execution pipelines for 7-DOF manipulators in industrial automation cells. Implemented C++ and Python software stacks for pick-and-place, welding applications. Delivered technical training sessions to Yaskawa engineering teams on ROS 2 integration, modern robotic software architectures, and AI-driven approaches to industrial automation. Presented *MotoROS2 – Yaskawa Motoman Interface for ROS 2 Applications* at the ROS-Industrial Conference 2023 in Bordeaux (Jul 2023), co-located with ROSCon Fr and RoboCup 2023.

Founder and Director **Jul 2020 – Sep 2023**
Buchholz Robotics, Norway

Independent robotics R&D consultancy and open-source development. Created and published three professional online robotics courses through The ConstructSim platform (Machine Learning for Robotics; Reinforcement Learning for Robotics; Behavior Trees for ROS 2), reaching thousands of engineers globally. Contributed open-source GazeboSim simulation environments for marine surface and underwater robotics. Authored 100+ technical articles on robotics, autonomous systems, and AI for a global engineering audience. Delivered contract robotics consulting across software architecture, ROS 2 integration, and autonomous systems design for Norwegian and international clients.

Senior Robotics Engineer – 3D Vision and Perception **Mar 2022 – Sep 2022**
Zivid, Norway

Developed robot guidance systems based on Zivid structured-light 3D cameras. Implemented point cloud processing, object recognition, and pose estimation pipelines for industrial bin-picking and precision assembly applications. Designed and validated robot-sensor extrinsic calibration workflows. Integrated Zivid perception systems with ROS 2 and industrial robot controllers (ABB, Yaskawa, UR). Characterised sensor performance under varying industrial lighting and surface conditions.

Principal Software Engineer (Technical Lead) **Jan 2018 – Jul 2020**
Canrig Robotics (Nabors Industries), Norway

Held decisive architectural authority over autonomous drilling robot development for deployment on active offshore rigs in harsh North Sea and Arctic conditions. Designed real-time motion control system, advanced path planners, safety-system and trajectory optimisers for drill-pipe handling in constrained three-dimensional workspaces, and computer vision pipelines for pipe detection and tracking. Developed dynamic models for load verification and impact detection, collision detection and avoidance systems, and applied deep learning to predictive maintenance and operational optimisation. Led software integration and commissioning on live operational rigs. Delivered knowledge-sharing sessions to multidisciplinary engineering teams on advanced control architectures, computer vision, and deep learning toolchains for robotics. Resulted in two granted US patents.

R&D Senior Engineer – Motion Control **Mar 2014 – Dec 2017**
ABB Robotics, Norway

Core team member for full-cycle development of three new industrial manipulator product lines, from mechanical design through motion control software to customer deployment. Contributed to the motion-control core of **RobotWare** (ABB's robot operating system) — software shared across the entire ABB manipulator range — and to **ABB RobotStudio** (offline programming and simulation environment). Developed rigid-body dynamics models, kinematic calibration methods, trajectory optimisation algorithms, and real-time embedded control software (C++) on ABB IRC5 controllers. Resolved motion-control challenges on customer production lines, primarily in the automotive industry (BMW, Volkswagen, Mercedes-Benz), including on-site testing, tuning, and performance trimming of deployed robots. Led development of SafeMove2 — ABB's human-robot collaboration safety system — including formal safety certification. Developed the Continuous Motion Rig (CMR) advanced motion control system for high-performance drilling operations. Designed and delivered technical training sessions to engineering teams across Norway, Sweden, and Germany on new robotic solutions, motion control architecture, and AI applications.

Senior Engineer – Subsea Control Systems **Mar 2012 – Feb 2014**
Equinor (Statoil), Norway

Operator-side technical expert for subsea control systems at Statoil (now Equinor), the field operator, overseeing subcontractors delivering deepwater production equipment. Led subsea control-system deliveries from factory acceptance through offshore commissioning into stable production, and diagnosed and resolved control-system faults on live installations. Engaged in new deliveries to the Snøhvit field and in control-system upgrades at Troll and Gullfaks. Oversaw system integration and hardware-in-the-loop testing of subsea control modules operating at full ocean depth in oil-filled pressure-rated housings, communicating over umbilical cables.

R&D Senior Engineer – Ship Electrical Power Systems **Jan 2011 – Feb 2012**
Rolls-Royce Marine, Norway
Developed hybrid electrical power systems and energy management architectures for dynamically positioned offshore vessels. Designed load distribution and power quality control algorithms for vessels operating under continuous wind and wave loading. Contributed to heave compensation control systems for offshore crane operations and to control architectures for subsea tidal energy generator systems. Integrated power electronics, switchgear, and energy storage with vessel dynamic positioning systems.

Senior Engineer – Subsea Control Systems **Apr 2008 – Dec 2010**
General Electric (GE Oil & Gas), Norway
Designed subsea control systems for offshore oil and gas production facilities. Developed pressure-rated electronic module architectures and embedded control software for subsea production and intervention equipment. Conducted hardware-in-the-loop simulation, system verification, and offshore field installation. Worked within international engineering teams on major North Sea subsea development projects.

Electrical Engineer (Marine) **Apr 2006 – Dec 2007**
Rysjedal Elektro AS, Norway
Electrical system installation, integration, and commissioning on commercial and offshore vessels. Responsible for power distribution, navigation electronics, and communication systems, in compliance with DNV and IMO marine standards.

Pedagogical Experience

Postgraduate Certificate in Teaching and Learning (PGCertTL) **2024**
Associate Fellowship of the Higher Education Academy (AFHEA)
Heriot-Watt University, Scotland

Teaching Assistant – Robotics Systems Science (*approx. 30 sessions*) **Sep 2023 – Dec 2025**
Lab supervision, tutorials, student support, and assessment contribution.
Heriot-Watt University, Scotland

MSc Project Supervisor – Baudouin Belpaire **2024**
Autonomous Waste Collection Pipeline on BlueBoat USV
Full supervision from system design to physical deployment.
Heriot-Watt University, Scotland

BSc Project Supervisor – Chen Xiangru **2025**
LLM-Driven Behaviour Trees for Autonomous Underwater Robotics
Student subsequently admitted to doctoral programme in Beijing.
Heriot-Watt University, Scotland

Hardware Project Co-supervisor – David Valente & Dean Rowlett **2023 – 2025**
Physical AUV construction: mechanical design, sensor integration (DVL, sonar, vision), and full ROS 2 control architecture. Both students now in senior professional positions (HWU National Robotarium; industry).
Ocean Systems Lab, Heriot-Watt University, Scotland

Creator – Online Course: Behavior Trees for ROS 2 **2022**
The ConstructSim (global professional robotics education platform)

Creator – Online Course: Reinforcement Learning for Robotics **2020**
The ConstructSim (global professional robotics education platform)

Creator – Online Course: Machine Learning for Robotics (ROS) **2020**
The ConstructSim (global professional robotics education platform)

Industrial Technical Training Delivery **2014 – 2023**
Designed and delivered technical training sessions for engineering teams at ABB Robotics, Canrig Robotics, and Yaskawa Robotics. Topics included advanced control system architectures, motion control, ROS 2 integration, computer vision, deep learning toolchains, and AI-driven approaches to industrial automation.

Project Acquisition and Coordination

UNITE – Multi-Robot Autonomous Inspection (Fugro) **2023 – 2025**
Lead software architect. Designed and built the complete autonomous ROS 2 software stack (ASV, AUV, 7-DOF manipulator) from concept to customer delivery.

HUME – HUman-machine teaming for Maritime Environments (SeeByte) **2023 – 2025**
Co-developer for autonomous multi-agent coordination system (ASV and multiple AUVs).

AQ^{2}UASIM Workshops – Commercial Outreach and Sponsorship **2024 – 2026**
Responsible for industrial fundraising; secured sponsorships from Norwegian and international maritime technology companies including Nortek, Waterlinked, Bluerobotics.

Administrative and Management Experience

Session Organiser – FFI Fagdag *Distribuert og skalerbar kampnær IKT*, Kjeller **2026**
Co-responsible, with Dr. Andreas Strand, for the session *Planleggingsstøtte med kunstig intelligens og simulering* (Planning Support with Artificial Intelligence and Simulation) at FFI's professional seminar for the Norwegian defence sector, May 2026.

Organising Committee Member – AQ^{2}UASIM Workshop, IROS 2026 **2026**
Responsible for commercial outreach, sponsorship, and programme coordination. The 2026 edition is listed as an official event of the UN Decade of Ocean Science for Sustainable Development (Ocean Decade) on the UNESCO/IOC Ocean Decade calendar.

Workshop Proposal Co-author – ROSCon UK 2025, Edinburgh **2025**
Co-authored the accepted proposal for the workshop *Teaching Robotics with ROS 2: Lessons, Platforms, and Perspectives* and contributed to its organisation, including speaker invitations and programme.

Organising Committee Member – AQ^{2}UASIM Workshop, ICRA 2025 **2025**
Responsible for commercial outreach, sponsorship, and programme coordination.

Founder and Director – Buchholz Robotics, Norway **2020 – 2023**
Full administrative, financial, and operational responsibility for independent robotics R&D and consultancy company.

Principal Software Engineer (Technical Lead) – Canrig Robotics **2018 – 2020**
Held decisive authority over software architecture and system solutions for autonomous drilling robot development teams.

Invited Talks and Presentations

Kunstig intelligens for beslutningsstøtte **May 2026**
(Artificial Intelligence for Decision Support)
FFI Fagdag, Norwegian Defence Research Establishment, Kjeller

Marine Robotics with SeeByte **Jun 2024**
Invited industry presentation, Heriot-Watt University, Edinburgh, Scotland

MotoROS2 – Yaskawa Motoman Interface for ROS 2 Applications **Jul 2023**
ROS-Industrial Conference 2023, Parc des Expositions de Bordeaux, France
(co-located with ROSCon Fr and RoboCup 2023)

Professional Service

ROS 2 Industrial Consortium – Member and Community Contributor	**2020 – present**
Open-Source Contributor: GazeboSim Marine Robotics Simulation Community	**2020 – present**
Author: Engineering Blog – 100+ technical articles on robotics and autonomous systems	**2018 – present**

Fellowships, Awards, and Patents

Associate Fellowship of the Higher Education Academy (AFHEA)	**2024**
US Patent (USPTO): Inventory System Design using Deep Learning	**2024**
US Patent (USPTO): Slip Wear Detection System using Deep Learning	**2022**
Golden Award from the Rector of Gdansk University of Technology	**2004**
Scholarship, Gdansk University of Technology (Computer Science)	**1998 – 2003**
Scholarship, Gdansk University of Technology (Economy)	**2002 – 2004**
Multiple National Medals, Track and Field – Decathlon	**1990 – 1998**

Publications

A complete, up-to-date list of publications is submitted as a separate document and is available online at Google Scholar.

Technical Skills

Programming: C++, Python, MATLAB
Robotics middleware: ROS 2
Optimal control and dynamics: Crocoddyl, Pinocchio, MoveIt 2
Simulation: GazeboSim, StoneFish, ArduPilot, Hardware-in-the-Loop (HILS)
Systems and deployment: Docker, Linux (embedded and desktop), Nvidia Jetson